2.8 The Quotient Rule — teacher key

Question: How do you differentiate one function divided by another?

Outline and pacing (42 min)

Tab	Priority	Minutes	If behind
start	1	2	never trimmed
warmup	1	4	never trimmed
predict	1	3	never trimmed
explore	3	8	skipped first
build	2	8	shrinks to 2 of 4
practice	1	8	never trimmed
check	1	4	never trimmed
Huddle h1	clock	minute 18	pulled in by the clock

Targets

- I can state the quotient rule.
- I can differentiate a quotient and find its slope at a point.
- I can use a table of values with the rule.

Predictions

- A student guesses that for $h = \dfrac{\sin x}{x^2}$ the derivative is $\dfrac{\cos x}{2x}$. At $x = 1$ that gives $+0.27$, but the graph of h is falling there. The guess is... **wrong: the sign disagrees with the graph.** The graph falls at x = 1, so the slope is negative. A positive value cannot be right.
- The curve $y = \dfrac{1}{x}$ for $x > 0$ falls as x increases. Its slope at $x = 2$ is... **negative.** The curve is falling, so the slope is negative.

Huddle script (answers in green)

Definitions

- **Quotient rule:** If h = f/g, then h? = (**g f? − f g?**) / **g²**.
- **Order:** The **bottom** times the derivative of the **top** comes first.
- **Shortcut:** If the bottom is one term, **simplify** first and use the power rule.

Worked example

Find h?(5) for h(x) = (x² + 1)/(x − 3).

- g f? − f g? = (x − 3)(2x) − (x² + 1)(1)
- At x = 5: 2 · 10 − **26** = **−6**
- Divide by g² = **4**: h?(5) = **−3/2**

Label the rule

1. The numerator is **bottom times top?** minus top times bottom?.
2. Order matters: **subtraction** is not commutative.
3. The denominator is the bottom **squared**.

Every level, with answers

Differentiate a quotient

Level 1. Differentiate with the quotient rule.

Answer: $\frac{-5}{(x-2)^2}$ Numerator: $(x-2)(1)-(x+3)(1)=-5$. So $\dfrac{-5}{(x-2)^2}$.

Level 2. Differentiate with the quotient rule.

Answer: $\frac{x^2+2x}{(x+1)^2}$ $\dfrac{(x+1)(2x)-x^2}{(x+1)^2}=\dfrac{x^2+2x}{(x+1)^2}$.

Level 3. Differentiate. Name f and g first.

Answer: $\frac{x^2\cdot e^x-2x^1\cdot e^x}{x^4}$ $\dfrac{x^2e^x-e^x\cdot 2x^1}{x^4}$.

Level 4. Differentiate. You do not have to simplify.

Answer: $\frac{x^5\cos(x)-5x^4\sin(x)}{x^{10}}$ $\dfrac{x^5\cos x-5x^4\sin x}{x^{10}}$.

Slope at a point

Level 1. Let $h(x)=\dfrac{x}{x+2}$. Find $h'(2)$.

Answer: 0.125 $h'(x)=\dfrac{(x+2)-x}{(x+2)^2}=\dfrac{2}{(x+2)^2}$, so $h'(2)=\dfrac{2}{16}=0.125$.

Level 2. Let $h(x)=\dfrac{x^2+5}{x-1}$. Find $h'(3)$.

Answer: -0.5 $h'(3)=\dfrac{(2)(6)-(14)(1)}{4}=-0.5$.

Level 3. A table gives $f(4)=-5$, $f'(4)=2$, $g(4)=10$ and $g'(4)=-6$. Let $h=\dfrac{f}{g}$. Find $h'(4)$.

Answer: -0.1 $h'(4)=\dfrac{(10)(2)-(-5)(-6)}{(10)^2}=-0.1$.

Level 4. A table gives $f(3)=8$, $f'(3)=-5$, $g(3)=10$ and $g'(3)=4$. Let $h=\dfrac{f}{g}$. Find $h'(3)$.

Answer: -0.82 $h'(3)=\dfrac{(10)(-5)-(8)(4)}{(10)^2}=-0.82$.

Simplify first

Level 1. Divide first, then differentiate.

Answer: 4 $\dfrac{4x^2+4x}{x}=4x+4$, whose derivative is 4.

Level 2. Divide first, then differentiate.

Answer: $4x$ $2x^2+7$ has derivative $4x$.

Level 3. Divide first, then differentiate.
Answer: $5 - \frac{2}{2x^{\frac{3}{2}}}$ $5x + 2x^{-1/2}$ has derivative $5 - \frac{2}{2}x^{-3/2}$.

Level 4. Divide first, then differentiate.
Answer: $5 + \frac{2}{x^3}$ $5x + 5 - 1x^{-2}$ has derivative $5 + \dfrac{2}{x^3}$.

Horizontal tangents

Level 1. Let $h(x) = \dfrac{x}{x^2+4}$ for $x > 0$. At which x is the tangent horizontal?
Answer: 2 $h'(x) = \dfrac{(x^2+4) - x(2x)}{(x^2+4)^2} = \dfrac{4-x^2}{(x^2+4)^2}$, which is 0 at $x = 2$.

Level 2. Let $h(x) = \dfrac{x}{x^2+4}$ for $x > 0$. At which x is the tangent horizontal?
Answer: 2 The numerator is $4 - x^2$, which is 0 at $x = 2$.

Level 3. Let $h(x) = \dfrac{x^2+25}{x}$ for $x > 0$. At which x is the tangent horizontal?
Answer: 5 The numerator is $x^2 - 25$, which is 0 at $x = 5$.

Level 4. Let $h(x) = \dfrac{e^x}{x^6}$ for $x > 0$. At which x is the tangent horizontal?
Answer: 6 $h'(x) = \dfrac{e^x x^5(x-6)}{x^{12}}$, which is 0 at $x = 6$ (e? is never 0).

Common misconceptions

Students say	Say instead
The derivative of f/g is f?/g?.	Test it: for sin x / x^2 at x = 1 the guess has the wrong sign. Use (g f? – f g?)/g^2.
Writing top·g? – bottom·f?.	Order matters. Start with bottom times the derivative of the top.
Forgetting to square the bottom.	The denominator is g^2, not g.
Using the quotient rule when the bottom is one term.	Divide first and use the power rule; it is shorter and safer.